EXPERIMENT #28

Timestamp: 20260323_131022 | Environment: Local

1. Experiment Parameters & Network Statistics

Parameter	Value
Number of sticks (N)	30
Stick length (a)	0.7
Domain size	1.0 × 1.0
Source region	x < 0.1
Drain region	x > 0.9
Total intersections	187
Has percolation	✓ Yes
# Percolating paths	1
Nodes before pruning	187
Nodes after pruning	13
Edges before pruning	186
Edges after pruning	12

2. Network Visualizations

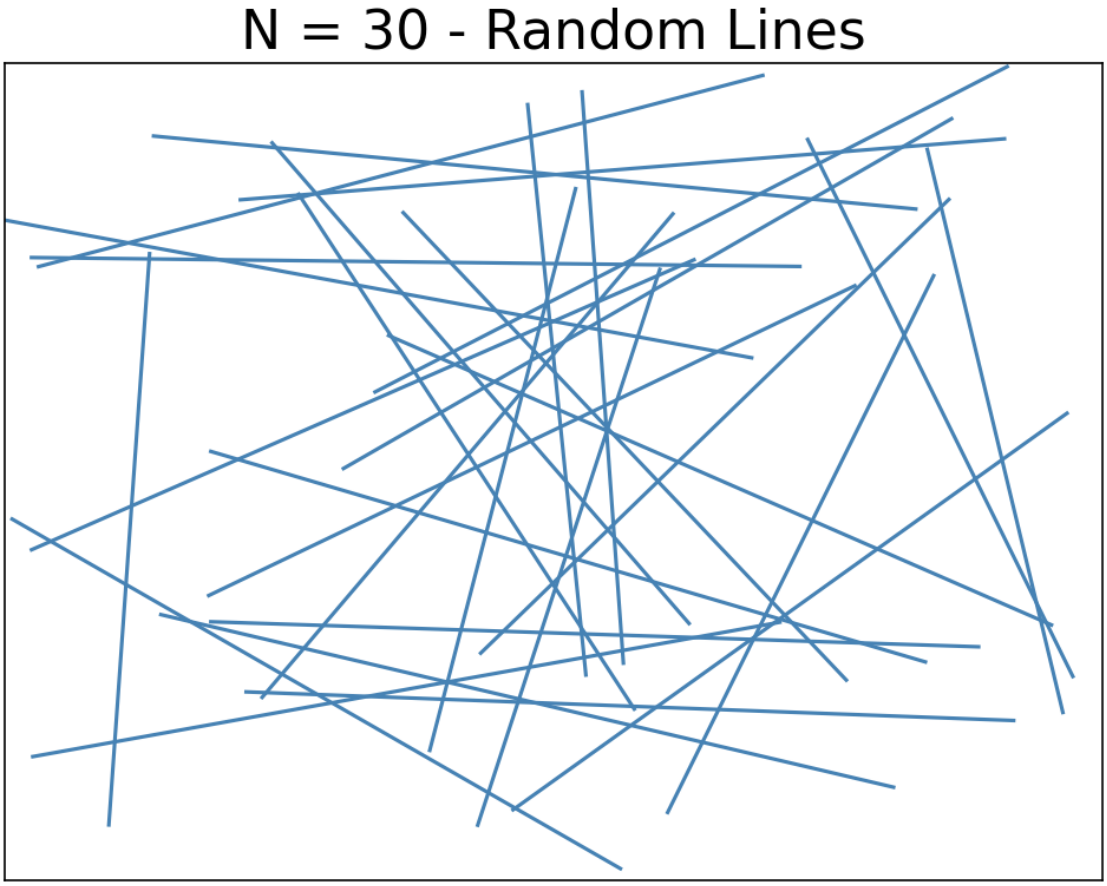


Fig. — 01 random lines N30

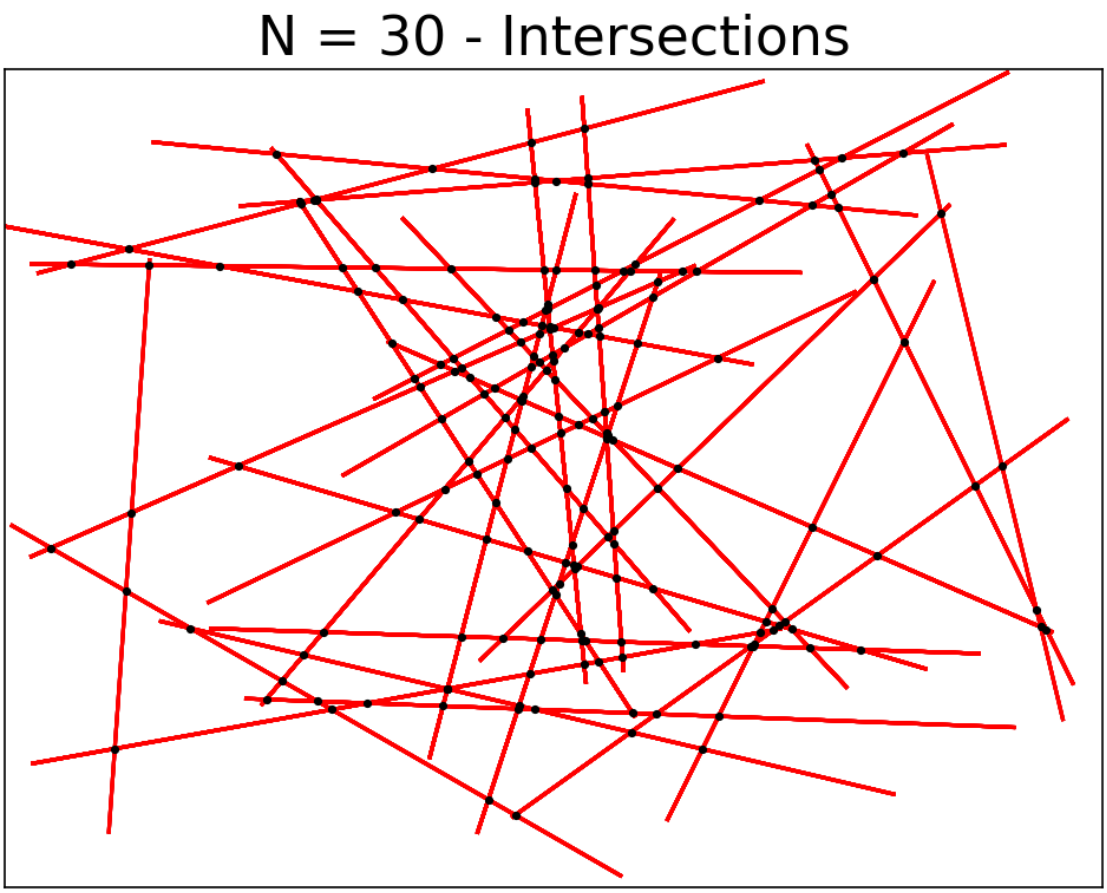


Fig. — 02 intersections N30

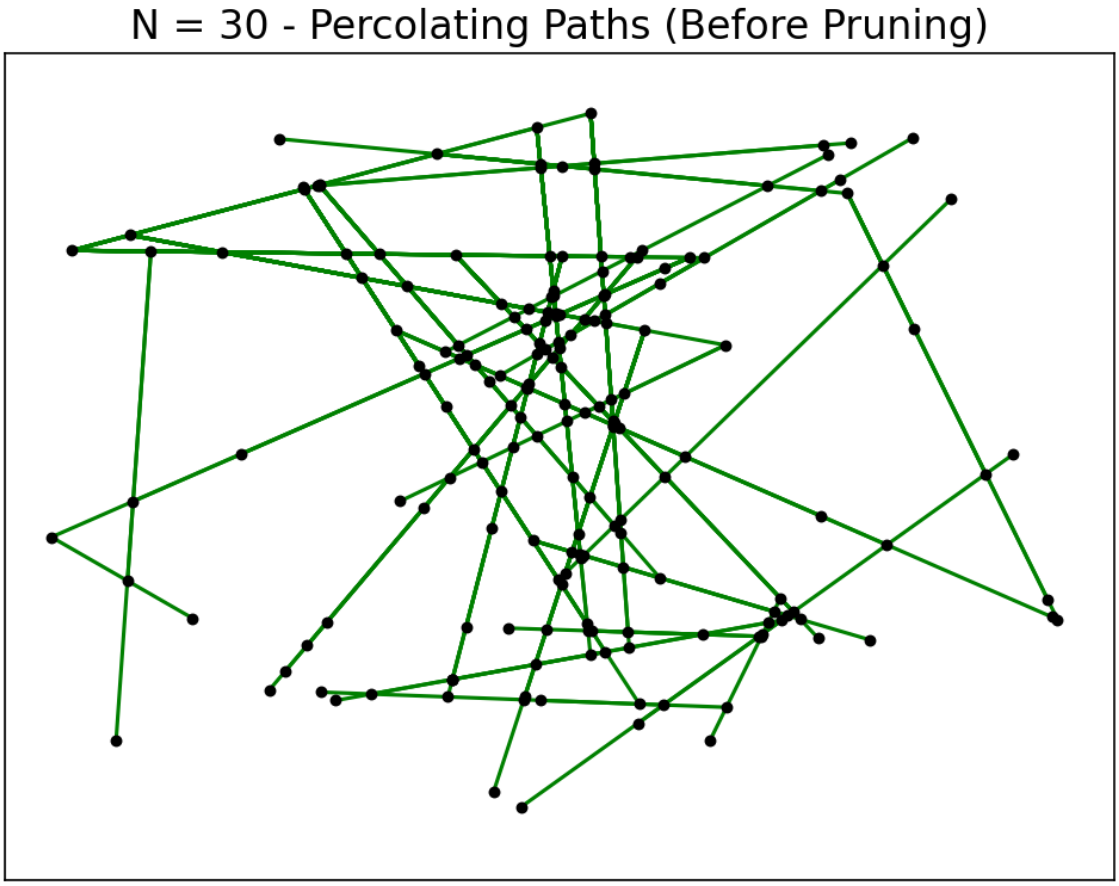


Fig. — 03 percolating paths before pruning N30

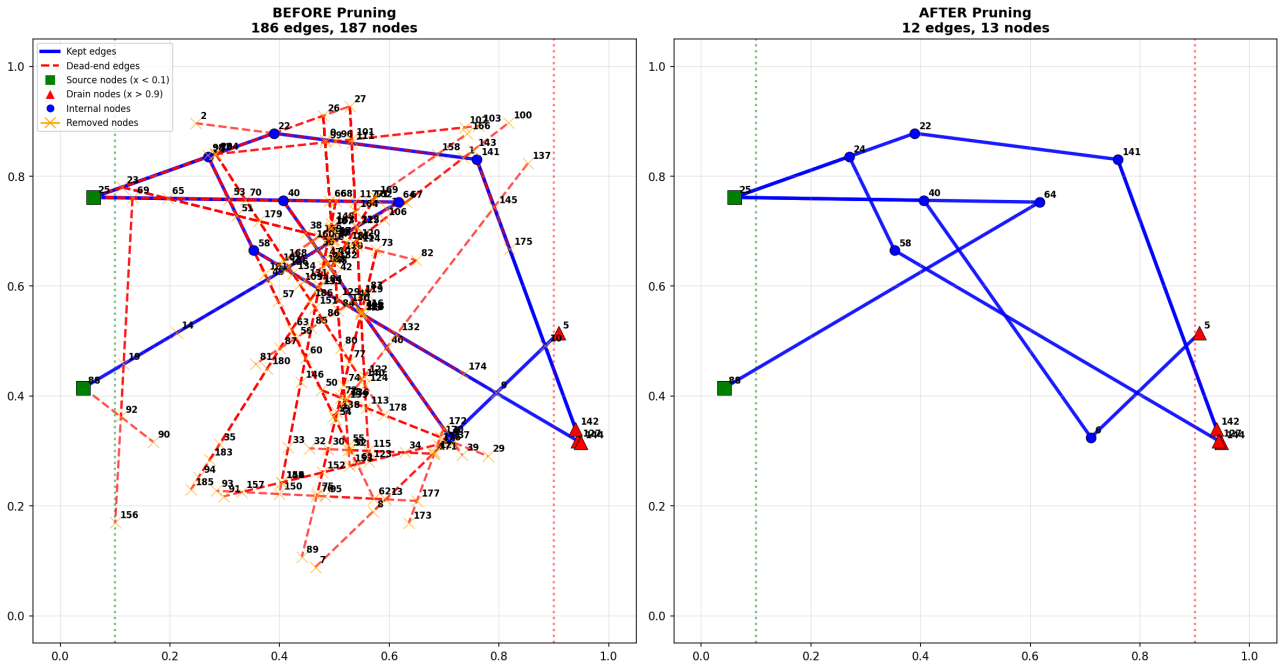


Fig. — 04 before after pruning N30

3. Edge Resistances (Pruned Network)

Edge #	Node 1	Node 2	Conductance (S)	Resistance (Ω)	Voltage Drop (V)
0	25	22	1.0000	1.0000	0.000000
1	25	24	1.0000	1.0000	0.000000

Edge #	Node 1	Node 2	Conductance (S)	Resistance (Ω)	Voltage Drop (V)
2	25	40	1.0000	1.0000	0.000000
3	25	64	1.0000	1.0000	0.000000
4	22	141	1.0000	1.0000	0.000000
5	24	58	1.0000	1.0000	0.000000
6	40	6	1.0000	1.0000	0.200000
7	64	88	1.0000	1.0000	0.000000
8	141	142	1.0000	1.0000	0.000000
9	141	144	1.0000	1.0000	0.000000
10	58	127	1.0000	1.0000	0.000000
11	6	5	1.0000	1.0000	0.800000

4. Impedance Calculation

Quantity	Value
Applied voltage (V)	1.00 V
Resistance per segment k	1 Ω
Calculated impedance R_eq	937.500000 Ω
Total current I	4.000000e-04 A

Impedance for different k (resistance per segment) values

k (Ω/segment)	R_equivalent (Ω)
1	0.937500
10	9.375000
100	93.750000
1000	937.500000
10000	9375.000000